Title: AI sensor and communication systems simulated in game engines (Godot/Unreal/Unity)
Desc.: Making a module/addon that has different AI's as plug-and-play components that can be used freely and run on local system. This should make NPC's easier and feel more living. These AIs are focused on sensor data expect one that is to use in communicating. Different types AI's planned: Machine vision, sound synthesizer, machine hearing, and LLM.

Research goals:
* (Main) Can AI systems be implemented as plug-and-play components while allowing them to run on local systems.
* How much resources must be used while using such AI's and how to optimize them.
* How do these AI's affect the environment/entity they are put in
* Can there be central logic unit that receives all information from the AI's and use it to navigate simulation
* Can this research be continued into real world robotics.

Tools used:
* Godot, Unreal, Unity (Game Engines)
* visual studio code (main coding)
* GitHub (version control)
* AI models (Gemma 4:e4b, and other open source that will be figured out later)

Timetable:

Name	Desc.	Date/deadline
Planed	Publish work plan	25-5-2026
Official Start	Lets do this.	1-6-2026
Foundation	Write methodology and related research	7-6-2026
The interesting part	Implement, test, and document. Doing the actual research.	1-7-2026
Finalize document	Ensure that everything is as it should be	14-7-2026
Realization	Sudden but planned realization that something went wrong.	15-7-2026
Recover	Fix everything in a panic and summer heat.	31-7-2026
The meeting	Present everything done during summer to the guide	??-8-2026
Redo	Lets redo this. Once more with feeling.	1-10-2026
Done	Finished, ready to be evaluated	1-12-2026
Graduate	I have did done it. I have become the software engineer in a 4,5 years.	6-1-2027
Kela money stops	Kela does pay no more. If the progress goes here, the situation just escalated to an awful level.	1-3-2027
"*Life happens*"	If everything falls apart, this is my absolute last deadline. This is the last plan that should **NOT** happen.	1-6-2027

The plan has been set. The path is lit. Now I only require the strength to follow it.

Related Works:

April 2, 2026, Lahiri Anusuya made an article about Alphabet Inc. Google Deepmind's new LLM models with family name Gemma 4. They are so efficient that they can be run on directly on user's own device. The models support more than 140 languages and due to its capability to be run on user's device, multi-step planning and autonomous actions are possible. (Lahiri, 2026)

A research called "*TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate*", introduces a methodology to Vector Quantization which aims to compress high dimensional vectors. Problem with compression usually is the loss of data but this research claims that their TurboQuant methodology is major improvement to preserving important vector data while compressing. The research provides formal proof that TurboQuant is close to the best information-theoretically achievable distortion rate which a Vector Quantizer could do. (Zandieh *et al.*, 2025)

This research called "*Development of Two Dimension (2D) Game Engine with Finite State Machine (FSM) Based Artificial Intelligence (AI) Subsystem*" made a web-based JavaScript and WebGL 2.0. game engine that had implemented FSM AI system to run behaviors. Vector based AI was just calculating distance from player and changing behavior based of that. FMS-based used FMS to implement behavior. The conclusion of this study was that FMS-based AIs can be used and performed 16.667% better than vector based AIs since FMS-based calculated only relevant logic while vector based calculated everything. (Adeniyi *et al.*, 2024)

References:
Adeniyi, A.E. *et al.* (2024) 'Development of Two Dimension (2D) Game Engine with Finite State Machine (FSM) Based Artificial Intelligence (AI) Subsystem', *Procedia Computer Science*, 235, pp. 2996–3006. Available at: https://doi.org/10.1016/j.procs.2024.04.283.

Lahiri, A. (2026) 'Google's Gemma 4 Brings AI Superpowers To Your Device', *Benzinga Newswires; Southfield* [Preprint]. Available at: https://ezproxy.cc.lut.fi/wire-feeds/googles-gemma-4-brings-ai-superpowers-your-device/docview/3324636798/se-2?accountid=27292 (Accessed: 25 May 2026).

Zandieh, A. *et al.* (2025) 'TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate'. arXiv. Available at: https://doi.org/10.48550/arXiv.2504.19874.